

Improved Quantum Approximate Optimization Algorithm for Low-Density Parity-Check Channel Decoding

Han Zeng, Fanxu Meng, Tian Luan, Xutao Yu, and Zaichen Zhang*

Quantum computing shows promise for 6G networks due to its parallel computing capabilities. In the context of the Noisy Intermediate-Scale Quantum era, the introduction of hybrid quantum-classical algorithms like Quantum Approximate Optimization Algorithm (QAOA) offer powerful solutions to many combinatorial optimization problems in 6G. This paper focuses on Low-Density Parity-Check (LDPC) channel decoding and proposes an improved QAOA algorithm assisted by the learning-to-learn strategy. We also investigate the parameter concentration phenomenon in QAOA-based LDPC decoding to assess the rationality. To evaluate effectiveness, a comprehensive numerical expression for the energy expectation of single-layer QAOA and propose indicators for transfer performance evaluation is provided. Based on simulation results, the similarity in parameter distribution across specific LDPC configurations is investigated. This similarity facilitates the transfer of training outcomes from smaller to larger-scale problems for optimization initialization, thereby avoiding the need for retraining. This approach offers insights and potential solutions for rapid, large-scale channel decoding in 6G networks, despite the current limitations of quantum hardware.

that conventional computing systems may be inadequate to meet these demands. An interesting area of study is the decoding strategy for Low-Density Parity-Check (LDPC) codes.^[2] LDPC codes provide exceptional error-correction capabilities for high-speed communication systems.^[3] However, achieving high performance with LDPC codes typically requires long code length, which can increase implementation and hardware complexities, necessitating significant memory resources, circuit area, and power consumption to maintain high data throughput rates.^[4]

One potential solution could be found in quantum computing, which relies on the principles of quantum mechanics to perform powerful calculations on problems which classical computers cannot handle.^[5,6] Quantum computing is expected to not only reduce computing time but also address the storage shortage problem associated with traditional hardware.^[7] Significant progress has been made in the development of current near-term noisy

intermediate-scale (NISQ) devices in recent years.^[8–10] The capacity to accommodate a greater number of qubits is steadily increasing, with several commercial and academic laboratories currently introducing NISQ equipment with over 100 qubits.^[11] Google's developers are optimistic about achieving a commercially viable quantum computer by 2029, while IBM aims to deliver the Condor quantum system, boasting more than 1000 qubits by the end of 2023.^[12,13] Both researchers and the industry are continuously advancing quantum algorithms and applications suitable for NISQ devices.

Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum-classical algorithm.^[14] It is a prominent quantum algorithm for NISQ devices, aiming to solve complex combinatorial optimization problems.^[15–17] According to Matsumine et al., QAOA can also be used to decode traditional channel codes on quantum processors.^[18] Researchers also found that the functionality of QAOA decoding is highly dependent on degree distributions.

Furthermore, Liu et al. propose a method to speed up the acquisition of quantum variational parameters for cost and mixer Hamiltonians using a learning-to-learn (L2L) framework.^[19] L2L framework is a meta-learning approach where an algorithm learns to adapt its parameters to different tasks that allow it to solve new problems more effectively.^[20–22] This approach could

1. Introduction

The development of 6G networks is poised to revolutionize the telecommunications industry, offering accelerated data transmission rates, improved connection reliability, and support for numerous interconnected devices.^[1] However, there are concerns

H. Zeng, X. Yu, Z. Zhang
Frontiers Science Center for Mobile Information Communication and Security
Southeast University
Nanjing 210096, P. R. China
E-mail: zczhang@seu.edu.cn

F. Meng
College of Artificial Intelligence Nanjing Tech University
Nanjing 211816, P. R. China

T. Luan
Yangtze Delta Region Industrial Innovation Center of Quantum and Information Technology
Suzhou 215100, P. R. China

X. Yu, Z. Zhang
Purple Mountain Laboratory
Nanjing 211111, P. R. China

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/qute.202300262>

DOI: 10.1002/qute.202300262

significantly reduce the time needed and the data exchanges between classical and quantum processors. However, the excessive number of layers in their circuits makes it challenging to prove the significant acceleration effect of L2L. In practical quantum circuits, implementation can be challenging and may introduce unintended errors.^[23]

In this study, we conduct simulations to investigate the parameter concentration phenomenon in QAOA for LDPC decoding issues. We temporarily exclude quantum noise from our analysis and focus on it as a future research direction. Additionally, we demonstrate the effectiveness of accelerating decoding using the L2L framework and its transfer capability. Our theoretical analysis of the parameter distribution of single-layer QAOA supports our proposal. We suggest that transfer can be achieved in LDPC encoding problems with similar QAOA parameter distribution patterns. This enables smaller-scale optimization procedures to be applied to larger-scale problems. This approach reduces training costs for neural networks and provides a potential solution for addressing larger-scale channel decoding problems as the advantages of quantum computing become more apparent with increasing problem scale.

The main contributions of this paper are summarized as follows:

- 1) We analyze the distribution of single-layer QAOA parameters for LDPC decoding, verifying the parameter concentration effect necessary for optimized initialization. To improve BER performance, instead of increasing QAOA depth directly for higher layers, we use a trained model for shallow QAOA and freeze these parameters while independently optimizing high-layer parameters, which also helps alleviate the impact of quantum noise.
- 2) We generalize the outcomes of the L2L framework and propose a comprehensive indicator to measure the transfer effect. Experiments show that this transfer capability enables smaller-scale training results to solve larger-scale problems, reducing training costs and enhancing data utilization efficiency.
- 3) We compare the bit-error rate (BER) performance of the L2L method with random initialization and the traditional belief propagation (BP) algorithm. Despite limitations due to quantum simulation and hardware scale, this comparison demonstrates the effectiveness and robustness of our approach.

The remainder of this paper is organized as follows. In Section 2, we present the system model and provide information on LDPC codes, QAOA, and the L2L framework. Section 3 discusses the parameter concentration phenomenon when using QAOA for channel decoding. We provide a theoretical derivation to test the feasibility of this framework, along with the possibility of transfer. In Section 4, we present simulation results along with complexity analysis. Finally, Section 5 concludes the paper and offers directions for future research.

2. System Model

2.1. LDPC Codes and Channel Model

The channel encoding and decoding process follow a fundamental principle: the channel encoder converts the input digital signal sequence $\mathbf{m}$ into the output digital sequence $\mathbf{c}$.^[2] The digital sequence $\mathbf{c}$ passes through a digital modem, encounters noise in the channel, and then $\hat{\mathbf{c}}$ is obtained. Subsequently, the channel decoder performs error detection and correction aiming to obtain a digital sequence $\hat{\mathbf{m}}$ that is as similar as possible to the original $\mathbf{m}$.

As a type of linear code, the LDPC code is characterized by its sparsity, enabling efficient encoding and decoding algorithms that can be implemented in hardware or software.^[24] The design of LDPC codes starts with constructing a parity check matrix $\mathbf{H}$, which is then used to determine a generation matrix $\mathbf{G}$ for subsequent encoding. The parity check matrix $\mathbf{H}$ can be divided into two parts:

$$\mathbf{H} = [\mathbf{Q}_{c \times m} \quad \mathbf{P}_{c \times c}] \quad (1)$$

Then calculate:

$$\mathbf{W} = [\mathbf{P}_{c \times c}^{-1} \quad \mathbf{Q}_{c \times m}] \quad (2)$$

The generation matrix $\mathbf{G}$ then can be presented as:

$$\mathbf{G} = [\mathbf{I}_{m \times m} \quad \mathbf{W}] \quad (3)$$

where $\mathbf{I}_{m \times m}$ is an identity matrix and the code words $\mathbf{v}$ can be obtained by $\mathbf{v} = \mathbf{G}\mathbf{u}$, assuming $\mathbf{u}$ represents the information bits.

In this paper, we consider a linear channel model with an (n, k) LDPC code in which the generator matrix is presented as $\mathbf{G}$, where n and k are the code length and information bit length respectively.

$$\mathbf{y} = \mathbf{x} + \mathbf{w} \quad (4)$$

where $\mathbf{w}$ is the additive white Gaussian noise (AWGN) with a zero mean and variance of σ^2 . Based on the binary-phase shift-keying (BPSK) principles, $\mathbf{v} = \mathbf{G}\mathbf{u}$ is mapped to $\{-1, 1\}^N$ and $x_i = (-1)^{v_i}$.

2.2. QAOA for Solving Channel Decoding Problems

QAOA aims to solve combinatorial optimization problems using quantum computing. To achieve the optimization, QAOA relies on two main components: the mixing Hamiltonian H_M and the cost Hamiltonian H_C . The mixing Hamiltonian uses a set of Pauli X gates to generate a superposition of solutions, while the cost Hamiltonian encodes the problem to be solved in the form of an objective function.

To present this cost Hamiltonian in the quantum circuit, we have:

$$U(C, \gamma) = \exp(-i\gamma H_C) \quad (5)$$

where $0 \leq \gamma \leq 2\pi$.

Similarly, the mixing Hamiltonian can be presented as:

$$U(B, \beta) = \exp(-i\beta H_M) = e^{-i\beta \sum_i \sigma_i^x} \quad (6)$$

where $0 \leq \beta \leq 2\pi$.

During the iteration, the parameters γ and β are updated as follows:

$$|\gamma, \beta\rangle = U(B, \beta_p) U(C, \gamma_p) \cdots U(B, \beta_1) U(C, \gamma_1) \quad (7)$$

where p denotes the circuit depth.

$$U(\gamma, \beta) = e^{i\beta_p H_M} e^{i\gamma_p H_C} \cdots e^{i\beta_1 H_M} e^{i\gamma_1 H_C} \quad (8)$$

Then the cost function can be presented as follows:

$$F_p(\gamma, \beta) = \langle \phi | U^\dagger(\gamma, \beta) H_C U(\gamma, \beta) | \phi \rangle \quad (9)$$

To complete the channel decoding process by QAOA, the problem needs to be encoded into cost Hamiltonian based on ML function:^[18,19,25]

$$\text{argmin} \sum_{v=1}^N (\gamma_v - (-1)^{x_v})^2 \quad (10)$$

Notice that $(-1)^{x_v}$ can be presented by quantum Pauli Z operator, the cost Hamiltonian is:

$$H_C = - \sum_{v=1}^N C_v = - \sum_{v=1}^N \gamma_v \prod_{k \in L_v^c} Z_k \quad (11)$$

where L_v^c is a set of nonzero-element indices in the v th column of the generator matrix $\mathbf{G}$, that is:

$$L_v^c = \{k : [\mathbf{G}]_{k,v} = 1\} \quad (12)$$

As we have mentioned in the channel model section, γ_v is the signal we receive from the channel and can be described as the BPSK symbol that has been corrupted by AWGN.

2.3. L2L Framework

L2L framework is a meta-learning approach that aims to enable an agent to automatically learn the best learning strategy for a given task. Its purpose is to minimize the information exchanges between a classical computer and a quantum processor when adjusting the trainable parameters in a quantum circuit. This framework chooses a long short-term memory (LSTM) recurrent neural network (RNN) to process sequence data where it accepts input vectors, produces output vectors, and retains some data in memory for future use.^[20,26]

When applying the L2L framework to QAOA, the optimization goal is to offer effective and reasonable parameter initialization suggestions. Suppose that at the previous time step, the parameters sent to the QAOA circuit is θ_{t-1} , the cost function result from

QAOA evaluated on the quantum computer is γ_{t-1} and the hidden state of the classical network is h_{t-1} . Treat the LSTM block as a black box, and after receiving these values, this block will adjust the internal weights ϕ and the new guess θ_t will be fed back to the circuit so the current cost function value γ_t would be obtained. The process of iterative updating can be presented as:

$$h_{t+1}, \theta_{t+1} = \text{RNN}_\phi(h_t, \theta_t, \gamma_t) \quad (13)$$

By repeating this training iteration, the weights in the recurrent neural network will be continuously adjusted and optimized to minimize the loss function of QAOA. After a few iterations, the system can give a good initialization suggestion. The schematic diagram of the L2L framework is shown in **Figure 1**.

Researches have explicitly presented the relationship between the scale and parameters of the QAOA structure, where the number of variational parameters in these ansatz depends solely on the number of alternating operators and not on the system size.^[27,28] This approach enables training RNN models on smaller QNN (Quantum neural network) instance sizes for specific problems, which can then be tested for transfer capability on larger QNN instance sizes.^[29]

To optimize the RNN weights ϕ , a suitable meta-learning loss function $\mathcal{L}(\phi)$ should be selected based on the distribution of a particular instance. In this paper, based on the distribution of QAOA parameters of channel decoding for LDPC codes, we choose the loss function as follows:

$$\mathcal{L}(\phi) = \sum_{t=1}^n \xi^{n-t} \gamma_t(\phi) \quad (14)$$

where n is the number of consecutive calls in the L2L framework, ξ is the decay coefficient and γ_t denotes the Hamiltonian cost function from the time step t corresponding to the weights ϕ . The exponentially decaying loss function implies that the output of the LSTM cell at later time steps will contribute more to the loss function.

3. Improved QAOA Channel Decoding

3.1. Parameter Concentration Phenomenon in QAOA

In this section, we aim to analyze the distribution of QAOA parameters in channel decoding across different LDPC encoding configurations. We apply the K-Means algorithm, a popular unsupervised machine learning algorithm for data clustering, to show the similarities of optimal parameters.

To assess the degree of parameter concentration, we utilized the silhouette score, which provides an effective means for evaluating the overall performance of a clustering algorithm on a given data set. This score, ranging from -1 to 1, indicates the quality of clustering, with higher values indicating better results. It is worth noticing that the deeper QAOA circuit is, the more parameters need to be optimized. Research has revealed a negative impact on parameter concentration with an increase in layer count on the same problem.^[28]

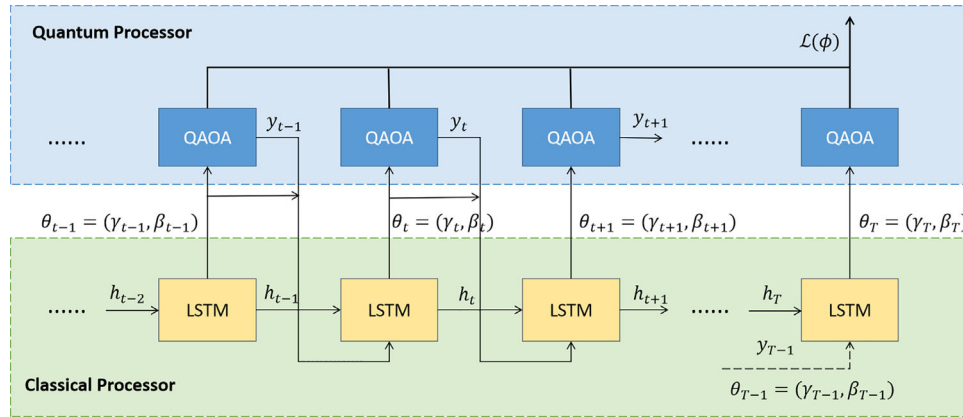


Figure 1. L2L framework.

Table 1. The detailed settings of simulations for single layer.

LDPC coding configuration	Qubits	Silhouette score	Layer	Centroids Coordinate
(12,2,3)	5	0.857	1	(-0.4673, 0.5041), (0.4672, -0.4907)
(12,2,4)	7	0.818	1	(-0.5572, 0.4312), (0.5499, -0.4131)
(12,3,6)	8	0.781	1	(-0.5765, 0.4117), (0.5563, -0.3759)
(16,2,4)	9	0.838	1	(-0.5327, 0.3640), (0.5252, -0.3648)
(18,2,3)	7	0.862	1	(-0.4340, 0.4341), (0.4271, -0.4361)
(20,2,4)	11	0.846	1	(-0.5074, 0.3262), (0.5124, -0.3308)
(20,3,4)	7	0.826	1	(-0.3485, 0.3446), (0.3446, -0.3109)
(20,3,5)	10	0.878	1	(-0.4258, 0.2241), (0.4280, -0.2164)

Table 2. The detailed settings of simulations for higher layers.

LDPC coding configuration	Qubits	Silhouette score	Layer	Centroids Coordinate
(12,2,3)	5	0.857	1	(-0.4673, 0.5041), (0.4672, -0.4907)
(12,2,3)	5	0.808	2	(-0.4362, 0.5392), (0.4338, -0.5583)
(12,2,3)	5	0.791	5	(-0.1924, 0.5422), (0.1955, -0.5583)
(12,2,3)	5	0.749	8	(-0.1350, 0.4596), (0.1375, -0.4534)
(12,2,3)	5	0.704	10	(-0.1158, 0.3991), (0.1139, -0.4169)

Limited by the ability of traditional computers to perform quantum simulations, we provide simulation examples for various regular LDPC codes, including (12,2,3), (12,2,4), and (16,2,4), among others, at an SNR of 5 dB. These examples allow us to observe the parameter distribution when using single-layer QAOA for channel decoding. Given the complexity of quantum noise, which may not be resolved by simple paragraphs, we will focus on quantum noise-free simulations for now. We have set the number of clusters to 2, and Table 1 displays the specific settings of these simulations, including the silhouette score of their parameter distribution.

Figure 2a–h visually displays the K-Means clustering results of the QAOA parameter distribution for specific LDPC encoding configurations. The parameters are represented by green points, and the centroids are marked in red. To demonstrate the simple symmetry based on the parameter distribution, we have marked the center of the centroids of the two clusters with blue dots. To investigate the impact of higher layers on the parameter distribution, in Figure 3a–d, we visualize the parameter distribution for different depths of QAOA, focusing solely on the parameters of the first layer for fairness in observation and comparison. Similar to Table 1, Table 2 shows the parameter concentration situation with higher layers.

The simulation results show that the QAOA parameter distribution generated by the LDPC decoding problem exhibits a concentration phenomenon. However, this effect diminishes as the number of layers increases, even when quantum noise is

ignored. While higher-layer QAOA offers enhanced expressibility and success probability, deeper circuits require more quantum resources and introduce additional computational complexity. Moreover, the susceptibility to quantum noise and errors increases with deeper circuits, leading to higher quantum error and decoherence rates.

Considering the feasibility of practical implementation and the concentration of parameter distribution, this paper starts with single-layer QAOA circuits for further analysis. As long as the number of layers in QAOA does not significantly weaken the parameter concentration effect, the results of parameter distribution as prior knowledge remain meaningful. Higher-layer configurations can be seen as extensions of the single-layer case, but their impact on quantum noise and error correction capabilities of actual quantum hardware requires further consideration, which will be our future research focus.

To further understand the numerical relationship between parameter distribution and the expected function in single-layer QAOA for channel decoding, we provide a more generalized theoretical analysis of the cost expectation landscape of single-layer QAOA.^[18]

First, the cost expectation in Equation (9) can be decomposed as:

$$F_p(\gamma, \beta) = \langle \phi | U^\dagger(\gamma, \beta) H_C U(\gamma, \beta) | \phi \rangle \quad (15)$$

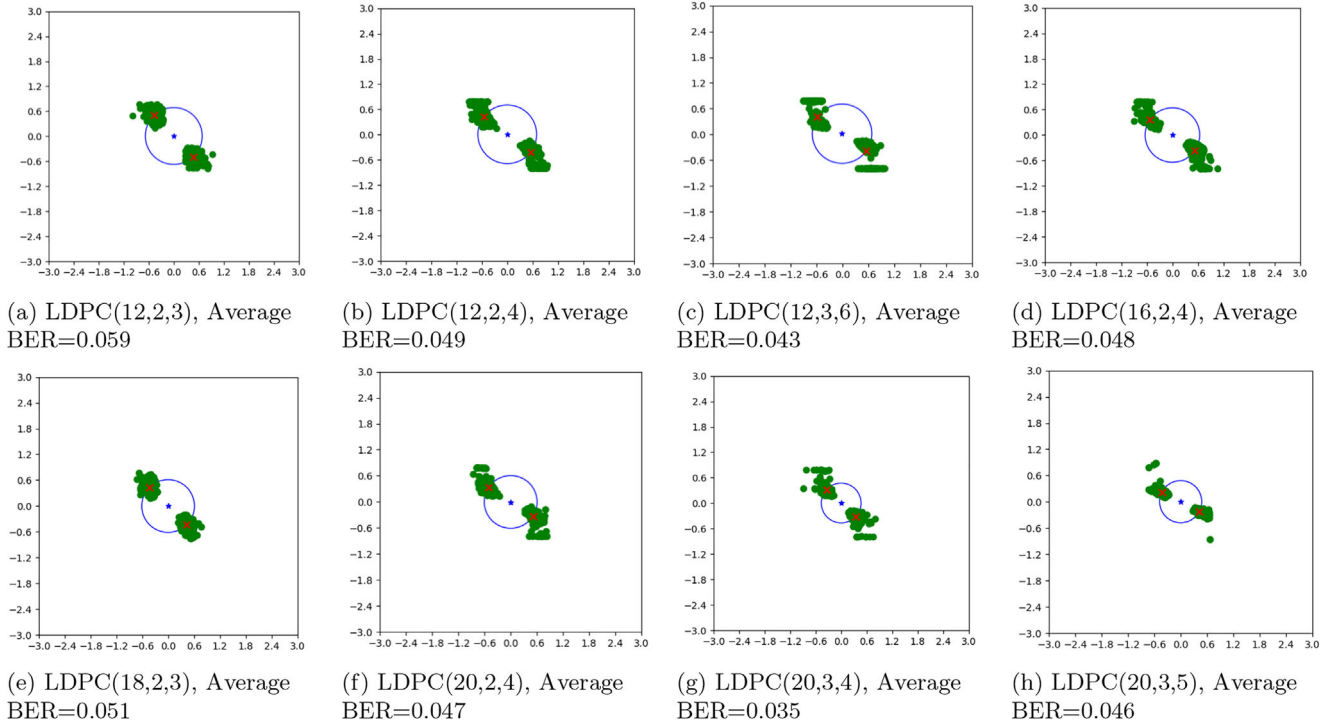


Figure 2. Different LDPC configurations and their QAOA parameter distribution (single layer).

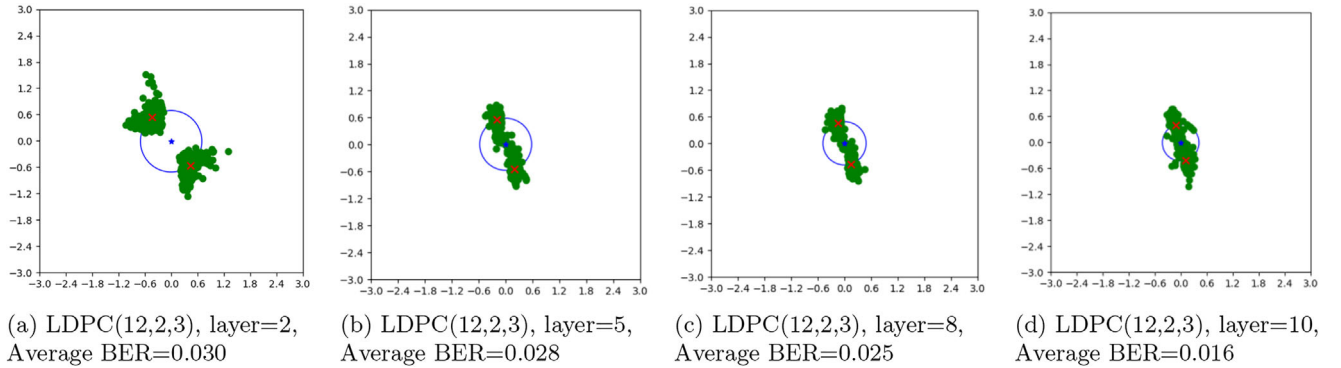


Figure 3. QAOA parameter distribution with higher layers.

Based on Equation (11), C_ν can be presented as:

$$C_\nu = -\gamma_\nu \prod_{k \in L_\nu^c} Z_k \quad (16)$$

Then we have:

$$U^\dagger(B, \beta) C_\nu U(B, \beta) = -\gamma_\nu \prod_{k \in L_\nu^c} (\cos(2\beta) Z_k + \sin(2\beta) Y_k) \quad (17)$$

Given the binary vector $\mathbf{b}_i$ ($0 \leq i < 2^{L_\nu^c}$) which has a bit length of L_ν^c and can represent the situation of the selected gate in each position. We assume that if we choose to apply a Z-gate at the i th position, then the corresponding vector index position will be set to 0. For example, suppose $L_\nu^c = 3$, then $\mathbf{b}_0 = [0, 0, 0]$ can be

mapped to the gate composition $M(\mathbf{b}_0) = Z_{k1} Z_{k2} Z_{k3}$. Similarly, $\mathbf{b}_4 = [1, 0, 0]$ can be mapped to $M(\mathbf{b}_4) = Y_{k1} Z_{k2} Z_{k3}$.

Letting ω_i be the weight of $\mathbf{b}_i$, according to the above rule, Equation (17) can be rewritten by the corresponding binary vector $\mathbf{b}_i$:

$$U^\dagger(B, \beta) C_\nu U(B, \beta) = -\gamma_\nu \sum_{i=0}^{2^{L_\nu^c}-1} (\cos(2\beta))^{L_\nu^c - \omega_i} * (\sin(2\beta))^{\omega_i} M(\mathbf{b}_i) = -\gamma_\nu \sum_{i=0}^{2^{L_\nu^c}-1} S_i M(\mathbf{b}_i). \quad (18)$$

where $S_i = (\cos(2\beta))^{L_\nu^c - \omega_i} * ((\sin(2\beta))^{\omega_i})$.

Then the original expression can be decomposed as:

$$U^\dagger(C, \gamma) S_i M(\mathbf{b}_i) U(C, \gamma) = \prod_v \left(\cos(\gamma_{Y_v}) \mathbf{I} - j \sin(\gamma_{Y_v}) \prod_{k \in L_v^c} \mathbf{Z}_k \right) \times \left(\cos(\gamma_{Y_{v'}}) \mathbf{I} - j \sin(\gamma_{Y_{v'}}) \prod_{k \in L_{v'}^c} \mathbf{Z}_k \right) \times S_i M(\mathbf{b}_i) \times \left(\cos(\gamma_{Y_{v'}}) \mathbf{I} + j \sin(\gamma_{Y_{v'}}) \prod_{k \in L_{v'}} \mathbf{Z}_k \right) \times \prod_v \left(\cos(\gamma_{Y_v}) \mathbf{I} + j \sin(\gamma_{Y_v}) \prod_{k \in L_v^c} \mathbf{Z}_k \right) \quad (19)$$

If $\prod_{k \in L_v^c} \mathbf{Z}_k$ and $M(\mathbf{b}_i)$ are non-commutable, then:

$$M(\mathbf{b}_i) \prod_{k \in L_v^c} \mathbf{Z}_k = - \prod_{k \in L_v^c} \mathbf{Z}_k M(\mathbf{b}_i) \quad (20)$$

$$U^\dagger(C, \gamma) S_i M(\mathbf{b}_i) U(C, \gamma) = S_i \times \left(\cos(2\gamma_{Y_{v'}}) \mathbf{I} + j \sin(2\gamma_{Y_{v'}}) \prod_{k \in L_{v'}} \mathbf{Z}_k \right) \times M(\mathbf{b}_i) \quad (21)$$

If $\prod_{k \in L_v^c} \mathbf{Z}_k$ and $M(\mathbf{b}_i)$ are commutable, then:

$$M(\mathbf{b}_i) \prod_{k \in L_v^c} \mathbf{Z}_k = \prod_{k \in L_v^c} \mathbf{Z}_k M(\mathbf{b}_i) \quad (22)$$

$$U^\dagger(C, \gamma) S_i M(\mathbf{b}_i) U(C, \gamma) = S_i \times \mathbf{I} \times M(\mathbf{b}_i) \quad (23)$$

Hence, according to $M(\mathbf{b}_i)$ and the generator matrix $\mathbf{G}$, the original equation can be simplified as follows:

$$U^\dagger(C, \gamma) S_i M(\mathbf{b}_i) U(C, \gamma) = S_i \times \prod_v^{p_i} \left(\left(\cos(2\gamma_{Y_{v'}}) \mathbf{I} - j \sin(2\gamma_{Y_{v'}}) \prod_{k \in L_{v'}} \mathbf{Z}_k \right) \times M(\mathbf{b}_i) \right) \quad (24)$$

where v' represent the column indexes where $\prod_{k \in L_v^c} \mathbf{Z}_k$ and $M(\mathbf{b}_i)$ are non-commutable. Suppose for each $M(\mathbf{b}_i)$, there are p_i eligible $v'(s)$, we use a binary vector $\mathbf{a}_j$ ($0 \leq j < 2^{p_i}$) which has a length of p_i and can represent the situation of the selected gate in each position.

For example, if choose $p_i = 3$, then $\mathbf{a}_0 = [0, 0, 0]$ can be mapped to the gate composition $M'(\mathbf{a}_0) = \mathbf{I}_{p1} \mathbf{I}_{p2} \mathbf{I}_{p3}$ and $\mathbf{a}_4 = [1, 0, 0]$ can be mapped to $M'(\mathbf{a}_4) = \mathbf{I}_{p1} \prod_{k \in L_{p2}^c} \mathbf{Z}_k \prod_{k \in L_{p3}^c} \mathbf{Z}_k$.

Similarly, the coefficient can be written as below:

$$S'_j[a_{j1}, a_{j2}, \dots, a_{jp_i}] = \prod_t^{p_i} S_t \quad (25)$$

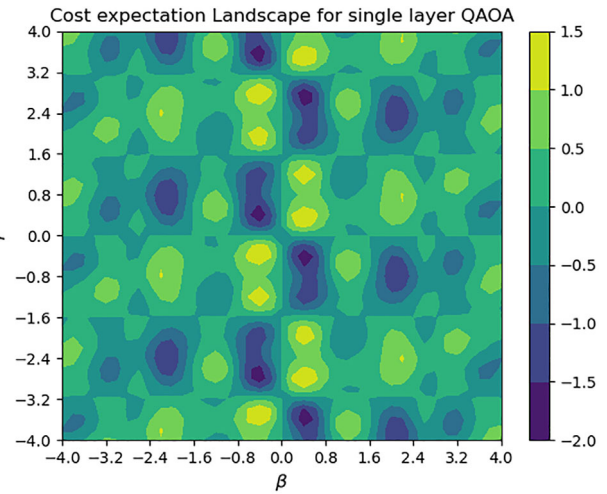


Figure 4. Cost expectation landscape for single layer QAOA.

where a_{jk} represents the k_{th} bit of a_j and

$$s_t = \begin{cases} \cos(2\gamma_{Y_{t'}}) & \text{if } a_{jt} = 0 \\ -j \sin(2\gamma_{Y_{t'}}) & \text{if } a_{jt} = 1 \end{cases} \quad (26)$$

For example, with $\mathbf{a}_4 = [1, 0, 0]$, $S'_{[1,0,0]} = \cos(2\gamma_{Y_{v'_1}}) * -j \sin(2\gamma_{Y_{v'_2}}) * -j \sin(2\gamma_{Y_{v'_3}})$.

Finally, we have:

$$F_p(\gamma, \beta) = \langle \phi | U^\dagger(\gamma, \beta) H_C U(\gamma, \beta) | \phi \rangle = - \sum_v \gamma_v \sum_{i=0}^{2^{L_v^c}} S_i \times \sum_{j=0}^{2^{p_i}} S'_j \text{Tr}[\rho_0 M'(\mathbf{a}_j) M(\mathbf{b}_i)] \quad (27)$$

where $\rho_0 = |\phi\rangle\langle\phi| = 2^{-n} \prod_{j=1}^n (\mathbf{I}_j + \mathbf{X}_j)$.

Here we give an example of regular LDPC (12,2,3) with its generator matrix:

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (28)$$

and its received signal is generated by random, we have $\mathbf{y} = [-0.446, 0.282, 1.000, 0.916, -0.462, -0.902, 0.824, -0.134, -0.512, -0.187, -0.122, -0.254]$.

Encoding the generator matrix and the received signal into a Hamiltonian, we plot the landscape based on the cost expectation value of the single-layer QAOA, resulting in **Figure 4**. By theoretically analyzing and substituting the generator matrix and received signal data into the previously derived cost function value expression, we obtain the corresponding landscape of the single-layer QAOA as shown in **Figure 5**. Comparing **Figures 4** and **5**, we briefly verify the correctness of the expectation function using this example. This expression allows us to theoretically observe

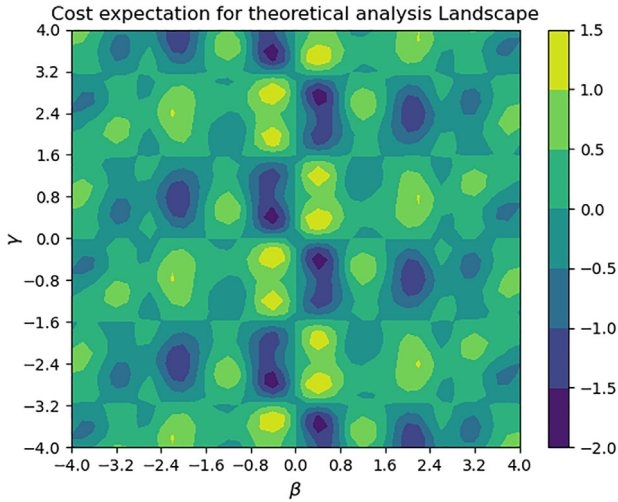


Figure 5. Cost expectation for theoretical analysis Landscape.

the concentration phenomenon in parameter distribution by utilizing the generator matrix and the received signal.

3.2. Overall Framework of Improved QAOA Channel Decoding

After observing the parameter concentration phenomenon in QAOA for channel decoding, we present the improved QAOA channel decoding framework in **Algorithm 1**. This framework includes training L2L LSTM cells on a training set, with the model weights fine-tuned during this process. Before applying the model to a new instance, an analysis determines if existing training results from the testing set can be suitably transferred. If transfer is deemed appropriate, the model weights are adjusted slightly based on the new instance, producing an optimized initialization suggestion for QAOA parameters after a few iterations.

3.3. Analysis of the Transfer Effectiveness

Given the substantial data requirements and complexity costs in training the internal weights of L2L, it's crucial to ensure that the trained framework demonstrates excellent generalization and transfer ability. Specifically, this article focuses on two aspects: first, the ability to transfer training outcomes from small-scale qubits to larger counterparts, and second, the ability to transfer training outcomes from higher SNR settings to lower ones.

To measure the transfer capability, we also introduce transfer effectiveness indicators. There are three main factors that influence the effectiveness of optimizing initialization:

- 1) Cost of Iterations: I_r and I_L
- 2) Initial cost value: C_r and C_L
- 3) Decoding errors: E_r and E_L

The subscript r represents the relevant indicators under random initialization, while the subscript L represents the relevant indi-

Algorithm 1 L2L-assisted QAOA channel decoding.

input : Training Data Set D_{train} , the loss function of LSTM cell L and the new instances D_{new}

output: The optimized initialization suggestion for parameters of QAOA

Training part:

Step1: Encode problems in D_t into Hamiltonian and get their cost function respectively.

Step2: Drop the cost function into L2L framework and start to learn the parameters of QAOA. Initialize the parameters and the weights in LSTM cell.

while Training do

Substitute parameters into the cost function and get the cost value;

Tuning the LSTM weights based on the cost value;

Give new parameter suggestions;

end

Step3: Obtain the trained weights.

Testing part:

Step1: Analyze whether the problems in D_n have cluster similarity with those in D_t .

if yes then

Go to Step 2 in Testing part;

else

Break and the training result is not suitable for this problem;

end

Step2: Tuning the weights slightly based on the given instance.

Step3: Return the optimized initialization suggestion for QAOA parameters.

cators under L2L optimization. Hence, we have the following indicator E_T to measure the transfer capability:

$$E_T = \alpha \cdot \frac{I_r - I_L}{I_r} + \beta \cdot \text{abs}\left(\frac{C_L}{C_r} - 1\right) + \gamma \cdot (E_r - E_L) \quad (29)$$

where α , β , and γ are weights set based on experience and satisfy $\alpha + \beta + \gamma = 1$. The higher E_T , the more effective the optimal results compared to that in the random initialization case.

4. Simulation Results And Analysis

4.1. Single-Layer QAOA with L2L Framework

Table 3 shows the basic settings of the L2L framework, QAOA circuit and traditional BP algorithm for simulation. The simulation settings include key parameters for the LSTM model (depth, learning rate, and optimizer), quantum simulation (library, optimizer, and step size), and specific settings for the quantum algorithms (maximum iterations for QAOA and BP). Monte Carlo simulations for BER calculation and coefficients for the Effective Index are also defined.

Table 3. The basic settings of the simulation.

Indicator name	Setting
LSTM depth	5
LSTM Learning rate	0.1
Decay coefficient	0.6
L2L training optimizer	Adam
L2L training cycles	5
Python library for quantum simulation	PennyLane ^[30]
Quantum optimizer	AdagradOptimizer
Quantum optimize stepsize	0.1
Maximum number of QAOA iterations	100
Maximum number of BP iterations	1000
Monte Carlo Times for Calculating BER	2×10^5
The coefficient used to calculate the Effective Index	$\alpha = 0.25, \beta = 0.25, \gamma = 0.5$

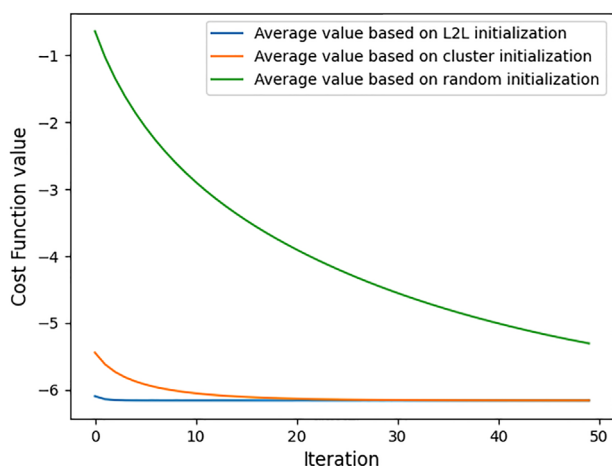


Figure 6. Cost expectation vs Iteration for LDPC(8,3,4).

4.1.1. Landscape and Cost Value

QAOA optimizes a cost function with parameters, and the algorithm's effectiveness also can be seen in the landscape of the optimal path chosen during iteration. We show these landscapes and paths for three initialization methods in single-layer QAOA: random, clustering, and L2L-recommended initialization based on two encoding configuration: regular LDPC(8,3,4) and regular LDPC(12,2,3). The training set includes 100 sets of channel data with the same LDPC constraints but different signal inputs. The testing set matches the LDPC encoding configuration of the training set, and we obtain 500 sets of different received signals. We record and average the cost function values for each parameter initialization method. The optimization paths for each method are also recorded and plotted in a landscape diagram where darker colors indicate lower energy.

As shown in Figures 6–9, both clustering-based and L2L-based parameter initialization methods result in lower energy compared to random initialization. This reduces the number of optimization iterations required for convergence with the same learning rate. These benefits are evident in the landscape diagram, as these initialization schemes often allow optimization

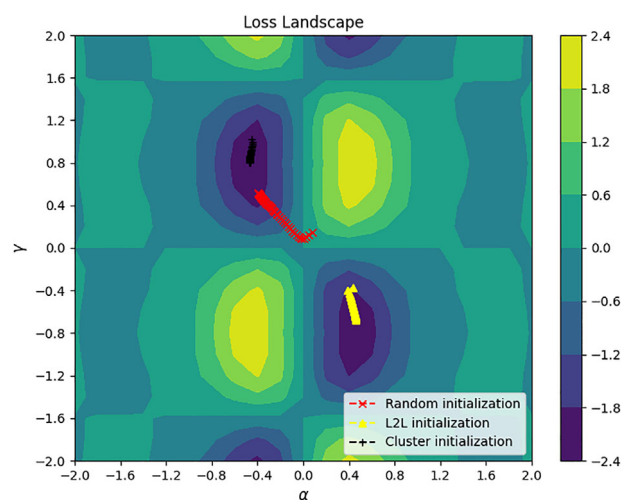


Figure 7. Energy Landscape and optimizing path for LDPC(8,3,4).

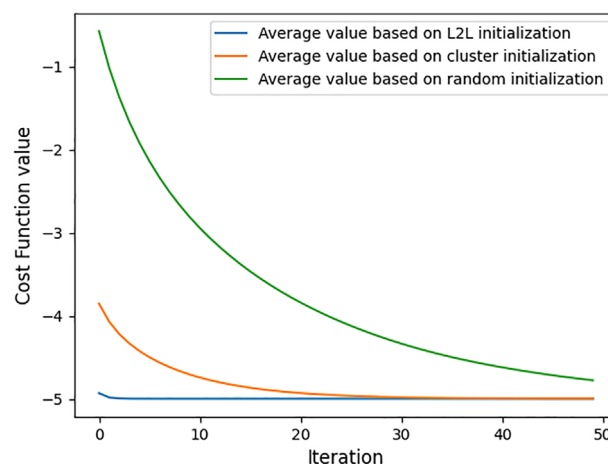


Figure 8. Cost expectation vs Iteration for LDPC(12,2,3).

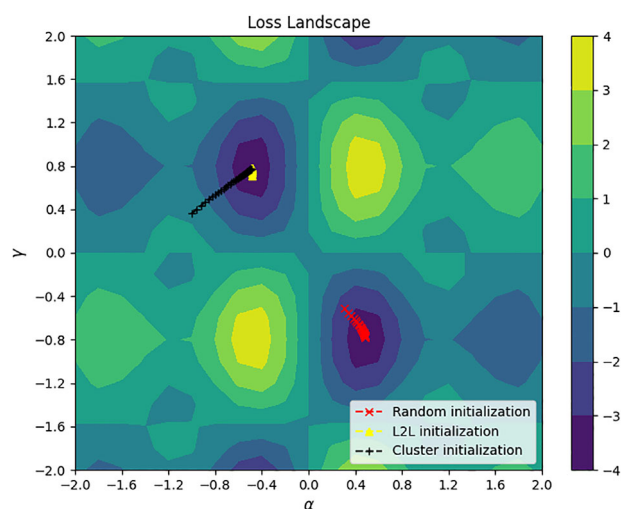


Figure 9. Energy Landscape and optimizing path for LDPC(12,2,3).

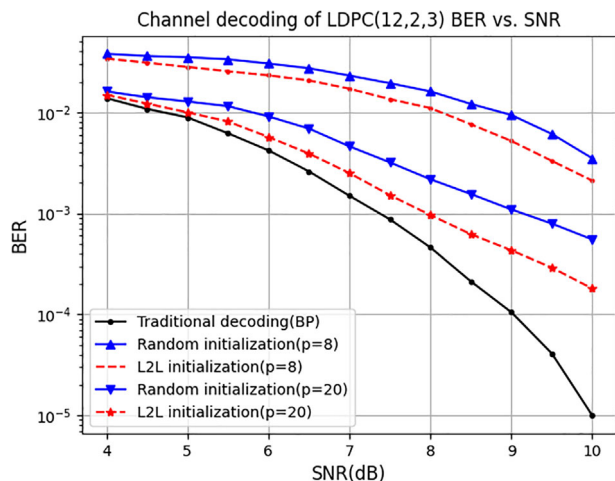


Figure 10. BER performance for LDPC(12,2,3).

to start from a lower energy level and reach the lowest energy point more efficiently within a certain range. Also, it is important to note that while the clustering-based scheme has lower initial training costs and shows similar effects to L2L in single-layer simulations, the concentration effect between parameters weakens as the number of QAOA layers increases. Compared to the L2L scheme, clustering may have limitations in terms of effectiveness and robustness.

4.1.2. BER of QAOA Channel Decoding for LDPC Codes

To demonstrate the effectiveness and robustness of the improved QAOA channel decoding scheme, we analyze the BER of the L2L initialization scheme in comparison to the random initialization pattern and conventional BP algorithm.^[31]

Based on the previous analysis of single-layer QAOA, we have chosen a relatively small number of layers to demonstrate the feasibility and effectiveness of LDPC decoding. Taking regular LDPC(12,2,3) as an example, we randomly select a generator matrix to confirm the encoding scheme and perform Monte Carlo simulations for different SNRs.

Although increasing the number of QAOA circuit layers can enhance BER performance, this improvement diminishes as the layers increase due to quantum noise and weakened parameter concentration effects. To address this, we optimize and initialize the model for shallow QAOA, freezing these parameters while independently optimizing the high-layer parameters. This approach helps mitigate the degradation of parameter concentration effects in higher-layer QAOA, leveraging the concentration effects of shallow QAOA for better performance. Additionally, fixing parameters is an effective strategy to alleviate quantum noise in higher-layer QAOA.

The initialization parameters from the L2L strategy outperform random initialization in terms of bit error rate, as shown in **Figure 10**. However, the BER performance of QAOA channel decoding still falls short of traditional decoding algorithms due to limitations in quantum hardware scale, error correction, and large-scale simulation capabilities. Nonetheless, the trend suggests that quantum decoding performance will eventually

match traditional algorithms with advancements in quantum hardware technology.

4.2. Transfer Exploration for Larger-Scale Cases

In this section, we will present simulation results of specific examples with QAOA to illustrate the transfer effectiveness of training outcomes that we discussed in the previous chapter. Additionally, we will provide measurable results for their respective transfer effectiveness indicators.

4.2.1. Larger Qubit Scale

As shown in **Figure 2**, at SNR = 5 dB, the parameter distributions of QAOA for LDPC(12,2,3), LDPC(18,2,3), and LDPC(16,2,4) decoding problems exhibit similarities. The corresponding Hamiltonians contain 5, 8, and 9 qubits, respectively. Based on this observation, we selected 100 sets of generator matrices conforming to the LDPC(12,2,3) encoding scheme and their corresponding randomly generated received signal data as the training set to train the weights in the L2L framework.

Table 4 presents the simulation results based on above LDPC encoding configuration. From a numerical perspective, using the L2L scheme for optimization initialization reduces the average number of iterations by half compared to random initialization, and improves BER performance by about 40%. The effective indicator obtained based on the number of iterations, initial cost value, and BER performance further proves the effectiveness of this transfer. Additionally, we compare the performance of optimization initialization results obtained from small-scale training results and directly using large-scale data for training, taking LDPC(18,2,3) as an example. In terms of BER performance, since both methods eventually converge through iterations, the difference between them is not significant. The main distinction lies in the average number of iterations, where the approach of directly using large-scale data for training shows some advantages. However, numerically, this advantage is not prominent.

Despite the larger qubit scale in the testing set compared to the training set, the similarity in parameter distribution allows for the good performance of the training results on the test set. Specifically, utilizing L2L initialization significantly reduces the number of QAOA iterations required, maintains similar learning rates, and improves error rates compared to random initialization.

4.2.2. Worse SNR Cases

From another perspective, considering the transfer ability of the L2L initialization scheme, as shown in **Table 5**, the training set utilizes relevant data with SNR = 10 dB. In the testing set, we investigate the impact of deteriorating SNR on transfer performance. From the simulation results, although there is some degradation in transferability with deteriorating SNR, compared to random initialization, the average number of iterations still maintains an advantage, with a relative superiority in BER performance of $\approx 30\%$ – 40% .

Table 4. Larger qubit scale case (SNR = 5 dB).

Data Set	Qubits	LDPC coding configuration	Ave. Random & L2L echo times	BER (Random vs. L2L)	Ave. L2L effective indicator
Training Set	5	(12,2,3)	-	-	-
Testing Set 1	8	(18,2,3)	82.7/40.7	0.0131/0.0077	2.59
Testing Set 2	9	(16,2,4)	87.8/46.8	0.0118/0.0074	2.65
Training Set	8	(18,2,3)	-	-	-
Testing Set 3	8	(18,2,3)	79.1/36.2	0.0129/0.0078	2.93

Table 5. Lower SNR cases.

Data Set	SNR [dB]	LDPC coding configuration	Ave. Random & L2L echo times	BER (Random vs. L2L)	Ave. L2L effective indicator
Training Set	10	(12,2,3)	-	-	-
Testing Set 1	8	(12,2,3)	67.8/32.1	0.0017/0.0012	2.44
Testing Set 2	6	(12,2,3)	69.3/34.4	0.0096/0.0079	2.27
Testing Set 3	4	(12,2,3)	72.9/39.2	0.0393/0.0241	2.13

While it is not feasible to cover all scenarios of received signals and generator matrices to eliminate randomness, the results in Tables 4 and 5 partially demonstrate the effectiveness and potential of this transfer. Moreover, although the L2L framework incurs some training costs, its transfer ability allows for fine-tuning using the existing training set instead of retraining, as long as the parameter distribution of the problem structure remains similar. This aspect demonstrates the robustness and efficiency of the L2L framework in optimizing QAOA parameter initialization.

4.3. Complexity Analysis

The complexity of quantum algorithms cannot be directly compared to that of classical algorithms due to their distinct computational models. Comparing their complexities is not straightforward, as they require different approaches for analysis and implementation.

For traditional decoding algorithms, we choose the Belief Propagation (BP) algorithm. Let n be the number of nodes in the graph and d_{max} be the maximum degree of any node in the graph. The time complexity of Belief Propagation depends on the number of iterations needed for convergence. Each iteration of the algorithm has a time complexity $(n \times d_{max}^2)$, since every node needs to exchange messages with their neighboring nodes, and each message requires computing a sum of products over all possible variable values. Therefore, the total time complexity of Belief Propagation is $O(t * n * d^2)$, where t is the number of iterations required for convergence.

To analyze the complexity of this quantum algorithm, we mainly divide them into two parts, namely the training cost based on the L2L framework and the cost required for QAOA to obtain expected values. Define N as the number of qubits, p as the circuit depth, n as the length of code words. In terms of the generator matrix G , define its sparsity as ρ :

$$\rho = \frac{\tau}{m \times n} \quad (30)$$

Table 6. Number of gates for p -layer QAOA circuit.

Name of Unitary Gates	Number of gates
Hadamard Gate	$3N$
CNOT Gate	$2[(p \times n - 1)N]$
Rotation-Z Gate	$N + n$
Total	$p[2(\rho n + 1)N + n]$

where $m \times n$ is the size of the matrix, and τ is the number of non-zero elements.

1) Cost based on the L2L framework

The complexity of this part mainly depends on the number of LSTM cells, the number of parameters to be optimized, and the size of the hidden layer. Define the number of training samples as S_t , the number of consecutive calls in each recurrent loop as C , and the number of training circles as N_c . The LSTM cell has an input setting of 1 cost value, $2p$ parameters and $2p + 2p$ hidden states where p depends on the depths on the depth of QAOA. The total number of trainable weights inside the LSTM modules can be presented as $W = (4p + 2) \times 8p$. Hence, the complexity of calculating and updating gradients can be presented as $O(2W)$.

To simplify the complexity analysis of the training process in the L2L framework, we express it as follows:

$$O(N_c * S_t * (C * (6p + 1) + 2W)) \quad (31)$$

The L2L framework replaces the traditional optimizer part of the QAOA algorithm, trading in small-scale learning costs for better transform performance instead of treating the problem completely new and optimizing from scratch.

2) The cost required for QAOA to obtain expected values

The complexity of QAOA is influenced by the number of quantum gates involved in the circuit. The number of each gate is shown in Table 6. For a given generator matrix, the p -layer QAOA circuit has $p[2(\rho n + 1)N + n]$ unitary gates. To obtain the cost

value, measurement operations need to be added at the end of the circuit. CNOT gates in a real quantum computer limit circuit depth. More CNOT gates mean longer execution times, higher decoherence risks, and potential accuracy loss and very deep QAOA on real hardware may be impractical. LDPC encoding's sparsity reduces CNOT gate needs, enabling tailored quantum circuits for LDPC code decoding to some extent.

5. Conclusion

In this paper, we have explored the parameter concentration phenomenon in QAOA-based channel decoding for LDPC codes. Our investigation has focused on the effectiveness of accelerating decoding using the L2L framework and highlighting its remarkable transform capabilities. Through simulations, we demonstrate that encoding with similar parameter distributions enables the use of smaller-scale training results for larger-scale problems. This transfer capability enables optimized parameter initialization with higher qubit scales or lower SNR, reducing training costs. To obtain better decoding performance, we suggest using the trained low-layer model to initialize and fix the optimization parameters while independently optimizing the higher-layer parameters, effectively alleviating quantum noise.

The future research will focus on investigating the robustness and effectiveness of the L2L framework in higher-layer QAOA circuits, considering the impact of quantum noise and addressing the challenges posed by practical quantum hardware.

Overall, our study contributes to the understanding of parameter concentration in QAOA based on specific problems and highlights the potential effectiveness improvement brought by exploring the transfer ability of the L2L framework assisting LDPC channel decoding. These results offer valuable insights and potential strategies to tackle channel decoding challenges on a larger scale, particularly within the context of 6G networks.

Acknowledgements

This work was supported by NSFC projects 61960206005 and 61871111, Jiangsu Key R&D Program Project BE2023011-2, the Fundamental Research Funds for the Central Universities 2242022k60001, and the Research Fund of National Mobile Communications Research Lab 2024A04.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

L2L strategy, LDPC, QAOA, quantum computing, transfer performance

Received: August 14, 2023
Revised: March 14, 2024
Published online: April 14, 2024

- [1] M. Z. Chowdhury, M. Shahjalal, S. Ahmed, Y. M. Jang, *IEEE Open J. Commun. Soc.* **2020**, 1, 957.
- [2] R. Gallager, *IEEE Trans. Inf. Theory* **1962**, 8, 21.
- [3] Z. Zhong, Y. Huang, Z. Zhang, X. You, C. Zhang, *IEEE Transac. Circuits Sys. II: Express Briefs* **2020**, 67, 3018.
- [4] J. Chen, A. Dholakia, E. Eleftheriou, M. Fossorier, X.-Y. Hu, *IEEE Transac. Commun.* **2005**, 53, 1288.
- [5] P. Botsinis, D. Alanis, Z. Babar, H. V. Nguyen, D. Chandra, S. X. Ng, L. Hanzo, *IEEE Commun. Surv. Tutor.* **2019**, 21, 1209.
- [6] B. Yan, Z. Tan, S. Wei, H. Jiang, W. Wang, H. Wang, L. Luo, Q. Duan, Y. Liu, W. Shi, Y. Fei, X. Meng, Y. Han, Z. Shan, J. Chen, X. Zhu, C. Zhang, F. Jin, H. Li, C. Song, Z. Wang, Z. Ma, H. Wang, G.-L. Long, *ArXiv*, **2022**, 2212.12372 [quant-ph].
- [7] F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, R. Barends, R. Biswas, S. Boixo, F. G. S. L. Brandao, D. A. Buell, B. Burkett, Y. Chen, Z. Chen, B. Chiaro, R. Collins, W. Courtney, A. Dunsworth, E. Farhi, B. Foxen, A. Fowler, C. Gidney, M. Giustina, R. Graff, K. Guerin, S. Habegger, M. P. Harrigan, M. J. Hartmann, A. Ho, M. Hoffmann, T. Huang, et al., *Nature* **2019**, 574, 505.
- [8] J. Preskill, *Quantum* **2018**, 2, 79.
- [9] J. W. Z. Lau, K. H. Lim, H. Shrotriya, L. C. Kwek, *AAPPS Bull.* **2022**, 32, 27.
- [10] H.-L. Huang, X.-Y. Xu, C. Guo, G. Tian, S.-J. Wei, X. Sun, W.-S. Bao, G.-L. Long, *Sci. China Phys., Mechan. Astron.* **2023**, 66, 250302.
- [11] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. Van Den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, A. Kandala, *Nature* **2023**, 618, 500.
- [12] Google Aims for Commercial-Grade Quantum Computer by 2029 - WSJ, <https://www.wsj.com/articles/google-aims-for-commercial-grade-quantum-computer-by-2029-11621359156>, accessed: July, **2023**.
- [13] IBM's roadmap for scaling quantum technology, **2021**, <https://research.ibm.com/blog/ibm-quantum-roadmap>, accessed: July, **2023**.
- [14] E. Farhi, J. Goldstone, S. Gutmann, *ArXiv*, **2014**, 1411.4028 [quant-ph].
- [15] F.-X. Meng, Z.-T. Li, Y. Xu-Tao, Z.-C. Zhang, *Quant. Sci. Technol.* **2022**, 7, 025002.
- [16] P. Botsinis, D. Alanis, Z. Babar, S. X. Ng, L. Hanzo, *IEEE Trans. Commun.* **2015**, 63, 3713.
- [17] P. Botsinis, D. Alanis, Z. Babar, H. V. Nguyen, D. Chandra, S. X. Ng, L. Hanzo, *IEEE Access* **2016**, 4, 7402.
- [18] T. Matsumine, T. Koike-Akino, Y. Wang, *ArXiv*, **2019**, 1903.02537 [cs, math].
- [19] B. Liu, T. Koike-Akino, Y. Wang, K. Parsons, *ArXiv*, **2022**, 2205.08611 [cs, eess].
- [20] G. Verdon, M. Broughton, J. R. McClean, K. J. Sung, R. Babbush, Z. Jiang, H. Neven, M. Mohseni, *ArXiv*, **2019**, 1907.05415 [quant-ph].
- [21] S. Khairy, R. Shaydulin, L. Cincio, Y. Alexeev, P. Balaprakash, *Proced. AAAI Confer. Artif. Intell.* **2020**, 34, 2367.
- [22] P. Chandarana, P. S. Veites, N. N. Hegade, E. Solano, Y. Ban, X. Chen, *ArXiv*, **2022**.
- [23] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, *Nat. Commun.* **2018**, 9, 4812.
- [24] C. E. Shannon, *Bell Sys. Tech. J.* **1948**, 27, 379.
- [25] J. Cui, Y. Xiong, S. X. Ng, L. Hanzo, *IEEE Transac. Commun.* **2022**, 70, 5386.
- [26] S. Hochreiter, J. Schmidhuber, *Neural Comput.* **1997**, 9, 1735.
- [27] L. Zhou, S.-T. Wang, S. Choi, H. Pichler, M. D. Lukin, *Phys. Rev. X* **2020**, 10, 021067.
- [28] C. Moussa, H. Wang, T. Bäck, V. Dunjko, *EPJ Quantum Technol.* **2022**, 9, 11.

- [29] S. C. Kak, *Adv. Elect. Phys.* **1995**, 94, 259.
- [30] V. Bergholm, J. Izaac, M. Schuld, C. Gogolin, S. Ahmed, V. Ajith, M. S. Alam, G. Alonso-Linaje, B. AkashNarayanan, A. Asadi, J. M. Arrazola, U. Azad, S. Banning, C. Blank, T. R. Bromley, B. A. Cordier, J. Ceroni, A. Delgado, O. Di Matteo, A. Dusko, T. Garg, D. Guala, A. Hayes, R. Hill, A. Ijaz, T. Isaacson, D. Ittah, S. Jahangiri, P. Jain, E. Jiang, et al., *ArXiv* **2022**, 1811.04968 [physics, physics:quant-ph].
- [31] R. Tanner, *IEEE Trans. Inf. Theory* **1981**, 27, 533.